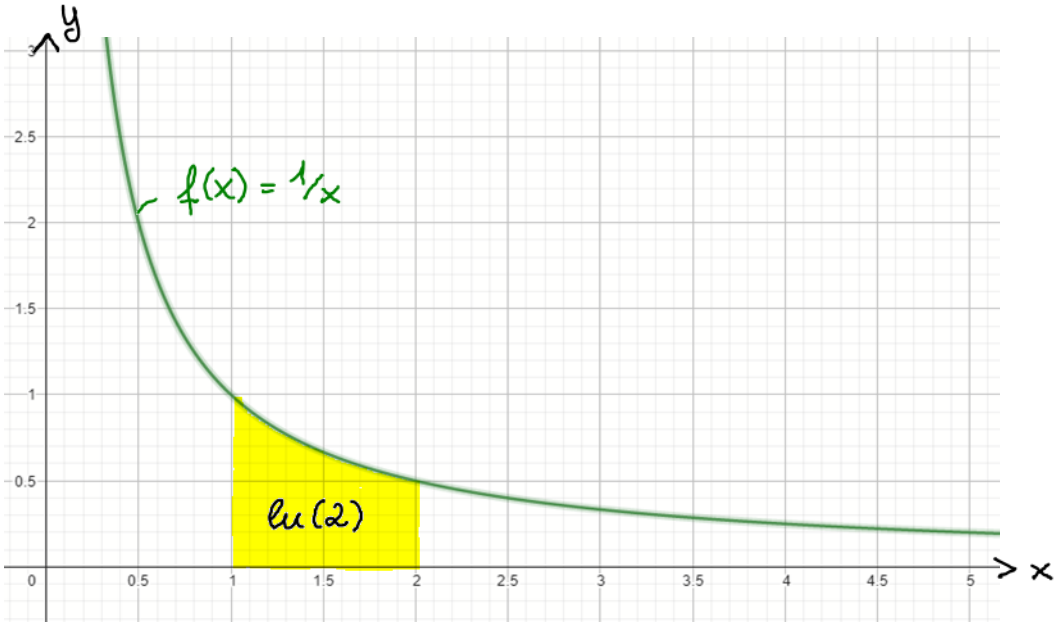
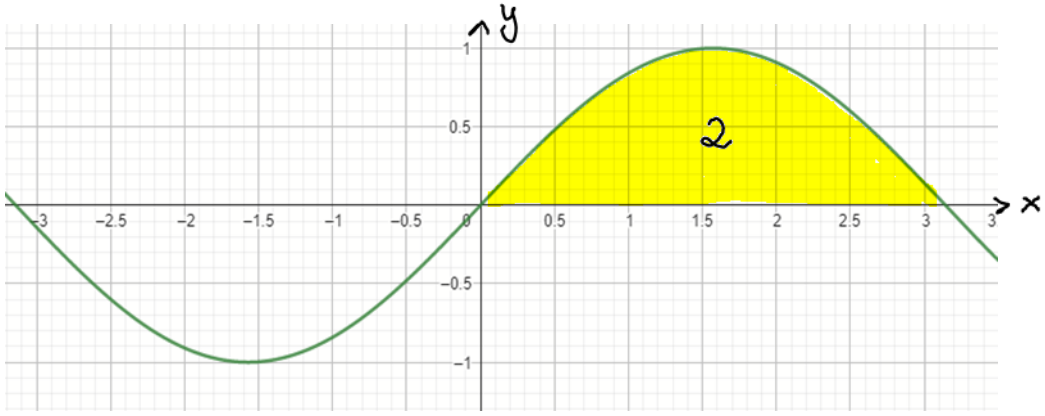
## Lösung: bestimmte Integrale

a) $$\int_1^2 \frac{1}{x}\,dx = \left[\ln(x)\right]_1^2 = \ln(2) - \ln(1) = \ln\left(\frac{2}{1}\right) = \ln(2)$$



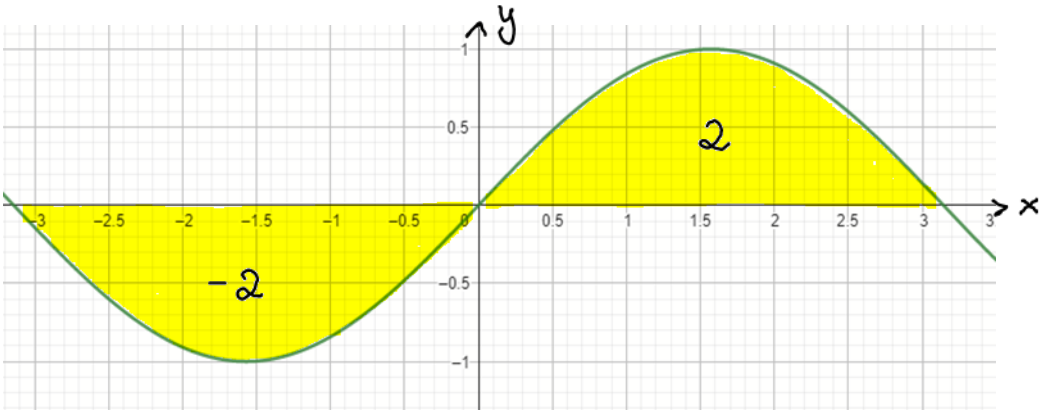

b) $$\int_0^{\pi} \sin(x)\,dx = \left[-\cos(x)\right]_0^{\pi} = -\cos(\pi) - (-\cos(0)) = -(-1) - (-1) = \underline{\underline{2}}$$



c) $\int_{-\pi}^{\pi} \sin(x)\,dx = [-\cos(x)]_{-\pi}^{\pi} = -\cos(\pi) - (-\cos(-\pi))$

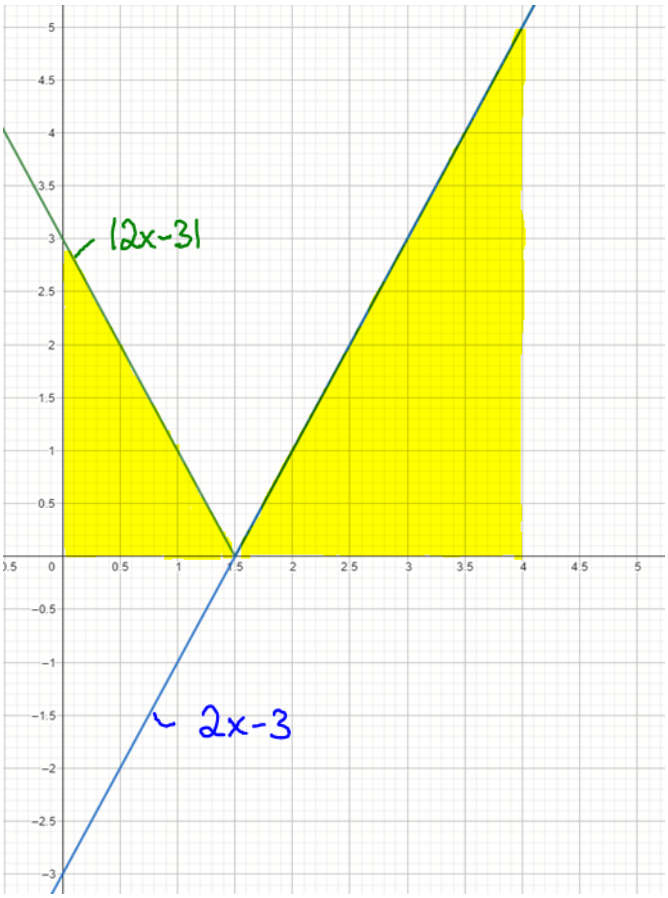
$= \cos(\pi)$

$= -(-1) - (-(-1)) = 1 - 1 = \underline{\underline{0}}$




Das bestimmte Integral einer ungeraden Funktion (d.h. ungerade Symmetrie) über ein symm. Intervall ist immer Null.

d) $\int_{0}^{4} |2x-3|\,dx = \ldots$




Unterscheide Bereiche in denen $2x-3$ positiv bzw negativ ist.

Nullstelle: $2x_0 - 3 = 0$

$2x_0 = 3$

$\underline{\underline{x_0 = 3/2 = 1{,}5}}$

$$\int_0^4 |2x-3|\,dx = \int_0^{3/2} \underbrace{|2x-3|}_{\leq 0}\,dx + \int_{3/2}^4 \underbrace{|2x-3|}_{\geq 0}\,dx$$

$$= \left(\int_0^{3/2} -(2x-3)\,dx\right) + \int_{3/2}^4 (2x-3)\,dx$$

$$= \left[-x^2+3x\right]_0^{3/2} + \left[x^2-3x\right]_{3/2}^4$$

$$= \left(-\left(\tfrac{3}{2}\right)^2 + 3\,\tfrac{3}{2} - (-0+0)\right) + \left(4^2 - 3\cdot 4 - \left(\left(\tfrac{3}{2}\right)^2 - 3\,\tfrac{3}{2}\right)\right)$$

$$= -\tfrac{9}{4} + \tfrac{9}{2} + 16 - 12 - \tfrac{9}{4} + \tfrac{9}{2} = 4 - \tfrac{18}{4} + \tfrac{18}{2}$$

$$= 4 - 4{,}5 + 9 = \underline{\underline{8{,}5}}$$